

COUNTING THE ACTIVE DEGREES OF FREEDOM OF A TRAINING RUN: WHAT A SINGLE SCALAR LOG CAN AND CANNOT MEASURE

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ABSTRACT

Within any window of training, the parameter trajectory of a neural network occupies a subspace far smaller than the parameter count. We treat the size of that subspace, the active dimension, as a count of components rather than a score, and ask how much of it survives the projection onto a single scalar log. The estimator is standard, delay reconstruction followed by a maximum likelihood intrinsic dimension; the contribution is knowing when its output is a count. On five systems of increasing realism, from a prescribed oscillation to a network trained on image data, whose active dimension is constructed and then verified by measurement, recovery is accurate to under one component up to eight directions on withheld ranks and seeds, and the ordering survives to twenty. Two diagnostics then decide from the series alone whether the value may be read as a count. Applied to grokking they place both standard experimental settings outside the admissible regime, for two different reasons which they identify correctly, while the estimate still separates generalising from non-generalising runs in four label-matched pairs. Measured directly from the trajectory covariance, the active dimension collapses to essentially one at the onset of generalisation and also falls in runs that never generalise.

1 INTRODUCTION

A neural network is trained in a parameter space of dimension P , but the trajectory the optimiser follows does not fill it. Within any window the parameter increments concentrate in a subspace of much smaller dimension, and the size of that subspace changes over the course of a run. We call it the *active dimension* of the window and define it in equation (2) as the participation ratio of the trajectory covariance, which counts the directions carrying comparable weight. It is therefore a count of components rather than a score: it states that one window excites four directions and the next excites one, on a scale that does not depend on the observer, the parameterisation or the task. Delay-embedding statistics return a plausible number whether or not it is such a count, and most of the work below goes into deciding which of the two one has.

It is not the dimension of the subspace the optimiser is confined to, which the parameterisation fixes (Li et al., 2018), nor the rank of the map from that subspace to the model’s outputs. Only the active dimension varies over a run, and section 3.1 shows the three to differ measurably.

Measuring it directly requires storing the trajectory in parameter space, which is expensive; a scalar log costs nothing, and a run already keeps several. We therefore ask how much of the active dimension survives the projection onto one such series. The instrument is standard, delay reconstruction (Takens, 1981; Sauer et al., 1991) followed by the pooled Levina–Bickel estimator (Levina & Bickel, 2004; MacKay & Ghahramani, 2005), and its hazard is that it returns a number when no invariant set exists at all: noise with a power-law spectrum yields a finite and reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989), and a decaying transient, which is what a converging run looks like, yields a large one. Section 4 therefore fixes a validation protocol before any result. Two of our five systems fail one of its controls, reproducing their reported result with no training taking place, and are kept in section 5.3 rather than removed.

Contributions.

1. We separate three quantities that are frequently identified with one another: the dimension of the subspace to which the optimiser is confined, the rank of the map from that subspace to the outputs, and the number of directions excited within a window (section 3.1). Only the third is estimated here, and it is measured rather than assumed throughout.
2. We evaluate the estimator on five systems of increasing realism, ending with a network trained on image data (section 5). It recovers the active dimension to 0.87 components up to eight directions, uncalibrated, on withheld ranks and seeds, and orders it to twenty.
3. We measure the conditions the absolute value requires (section 6) and validate two diagnostics that decide admissibility from the series alone.
4. Applied to grokking (section 7), those diagnostics place both published experimental settings outside the admissible regime; measured directly from the trajectory covariance, the active dimension collapses to essentially one at the onset of generalisation.

2 RELATED WORK

Delay reconstruction. Takens’ theorem (Takens, 1981), its prevalence form in terms of the box-counting dimension (Sauer et al., 1991) and its extensions to externally forced systems (Stark, 1999; Stark et al., 2003) license the reconstruction used here; empirical dynamic modelling applies the same machinery to ecological and physiological series (Sugihara & May, 1990; Sugihara et al., 2012). The lag and the embedding dimension are conventionally chosen by mutual information (Fraser & Swinney, 1986) and false nearest neighbours (Kennel et al., 1992; Cao, 1997), and the linear alternative to the whole procedure is the singular spectrum of the delay matrix (Broomhead & King, 1986), which we report throughout as PR_{delay} .

Intrinsic dimension, and when it is spurious. The dimension of the reconstructed cloud is estimated by the maximum likelihood estimator of Levina & Bickel (2004) with the pooling correction of MacKay & Ghahramani (2005), whose bias relative to the per-point average is known in closed form and is verified in section 5.1; the correlation dimension (Grassberger & Procaccia, 1983) and the neighbour-ratio estimator (Facco et al., 2017) are the standard alternatives. Two spurious regimes are documented: temporally adjacent neighbours inflate the estimate on short records (Theiler, 1986), and noise with a power-law spectrum produces a finite reproducible value (Osborne & Provenzale, 1989). The usual defence, a surrogate-data test against a linear-stochastic null (Theiler et al., 1992; Schreiber & Schmitz, 1996), answers whether nonlinear structure is present rather than how many components there are, and does not cover the regime that dominates optimisation logs: a deterministic transient has neither an attractor nor noise, and returns a large stable number where the true active dimension is one.

Dimension in optimisation. Random-subspace training (Li et al., 2018) fixes the available dimension and reports the smallest k at which a task is still solvable, a property of the task rather than of a window of a run; we borrow the parameterisation and measure a different quantity inside it. The closest existing statement to the active dimension is that gradient descent occurs largely in a small subspace spanned by the top Hessian eigenvectors (Gur-Ari et al., 2018), measured in parameter space from stored gradients, whereas the intrinsic dimension of data and of representations (Ansuini et al., 2019; Pope et al., 2021) is that of a cloud of points in activation space rather than of a trajectory; the participation ratio we adopt is the standard dimensionality measure for neural population activity (Gao et al., 2017). All of these need the weights, the gradients or the activations, and the nearest precedent for a scalar computed during training and then used to decide something is the gradient noise scale (McCandlish et al., 2018). The recurrent regime our estimate requires is not exotic, since full-batch descent typically operates at the edge of stability (Cohen et al., 2021), while mini-batch training near a minimum is well described as a stochastic process (Mandt et al., 2017) whose delay map embeds the state with its innovations (Stark et al., 2003).

Grokking. We reproduce the two settings in which delayed generalisation (Power et al., 2022) is usually studied: the regularised stochastic one of Liu et al. (2023), and the unregularised full-batch one of Gromov (2023) and Doshi et al. (2024), whose closed-form solution supplies both

label-matched target pairs and an analytic reference for the dimension of the learned representation. Progress measures and explanations for the transition are defined in weight or activation space (Nanda et al., 2023; Varma et al., 2023; Kumar et al., 2024; Liu et al., 2022). The work closest to ours reads the loss curve alone: Notsawo Jr. et al. (2023) classify hyperparameter configurations from spectral features of the first few hundred steps. We estimate instead a dynamical quantity with an independently measured ground truth, on systems where the answer is known before it is applied to runs where it is not.

3 PROBLEM STATEMENT

3.1 AVAILABLE, FUNCTIONAL AND ACTIVE DIMENSION

For a non-negative spectrum $\lambda_1, \dots, \lambda_n$ we write

$$\text{PR}(\lambda) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (1)$$

for its participation ratio, which counts the components carrying comparable weight: it equals m exactly when m of the λ_i are equal and the rest vanish, and it is invariant to a rescaling of the spectrum.

Let $\theta_t \in \mathbb{R}^P$ be the parameter vector at step t , and let the optimiser be confined, by construction or by the dynamics, to an affine subspace $\theta = \theta_0 + V^\top c$ with $V \in \mathbb{R}^{k \times P}$ orthonormal and $c_t \in \mathbb{R}^k$. Over a window $\mathcal{W}$ the *available* dimension is k ; the *functional* dimension is the rank of $\partial f(c)/\partial c$, where f collects the outputs on a fixed probe set, bounding how many available directions can change the computed function; and the *active* dimension is

$$d^{\text{act}}(\mathcal{W}) = \text{PR}(\text{spec Cov}\{c_t : t \in \mathcal{W}\}), \quad (2)$$

together with the same quantity computed on the increments Δc_t .

The three are distinct, and measurably so. In the system of section 5.4 the available dimension is 10 and the output Jacobian has rank 10; removing the preconditioner from the update rule leaves both unchanged, and the injected rank intact, while the active dimension falls from 7.99 to 6.43 under stochastic forcing, since anisotropic curvature redistributes the excitation across the available directions. Only the active dimension is estimated here, always against its measured value.

3.2 OBSERVATION MODEL AND ESTIMATOR

We observe a scalar series $x_t = \phi(s_t)$, with s_t the full optimiser state and ϕ fixed before the run, and reconstruct $y_t = (x_t, x_{t-\tau}, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a deterministic flow on a compact invariant set of dimension d this map is generically an embedding once $E > 2d$, so the reconstructed set is diffeomorphic to the original and carries its dimension (Takens, 1981; Sauer et al., 1991).

We estimate the intrinsic dimension of $\{y_t\}$ by maximum likelihood on nearest-neighbour distances (Levina & Bickel, 2004). With $r_1^{(i)} \leq \dots \leq r_m^{(i)}$ the distances from point i to its m nearest neighbours, subject to a Theiler exclusion W that forbids neighbours closer than W steps in time (Theiler, 1986), the local estimate is

$$\hat{d}_i = \left[\frac{1}{m-1} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}} \right]^{-1}, \quad (3)$$

and the two ways of combining the N local estimates differ. Averaging them, as Levina & Bickel (2004) propose, gives $\hat{d}_{\text{LB}} = N^{-1} \sum_i \hat{d}_i$. Pooling the likelihood instead, as MacKay & Ghahramani (2005) argue one should, averages the reciprocals:

$$\hat{d}_{\text{MG}}^{-1} = \frac{1}{N(m-1)} \sum_{i=1}^N \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}. \quad (4)$$

We write LB for the average of equation (3) and MG for the pooled form of equation (4). Since $\hat{d}_i^{-1}$ is a sum of $m-1$ exponential variates it is unbiased where $\hat{d}_i$ is not, so the two differ by the factor

Table 1: What a delay reconstruction of a scalar observation contains in each dynamical regime. Only the first admits a dimension.

regime	reconstructed object	its dimension
deterministic, recurrent	diffeomorphic copy of the r -torus	r
deterministic, transient	a one-dimensional curve in t	1, for every r
stochastically driven	state together with $E - 1$ innovations	full rank in $\mathbb{R}^E$

$(m - 1)/(m - 2)$, which section 5.1 confirms accounts for the whole gap between them; Appendix A states the algorithm.

Three further statistics accompany every MG value, because each can reproduce an apparent dimension signal without any geometry: the *roughness* $\text{std}(\Delta x)/\text{std}(x)$; the *linear participation ratio* PR_{delay} , equation (1) applied to $\text{Cov}\{\mathbf{y}_t\}$; and the *spectral participation ratio*, equation (1) applied to the power spectrum of x .

3.3 DYNAMICAL REGIMES

The embedding theorems require a deterministic flow on a compact invariant set the trajectory revisits. An optimiser near a solution may be in any of the three regimes of table 1, and only the first meets that requirement.

The third regime is what mini-batch gradient noise produces. Embedding theorems for stochastically forced systems exist (Stark, 1999; Stark et al., 2003), but what they embed is the state together with its noise history, a skew product rather than an r -dimensional attractor, and coloured noise alone yields a finite reproducible dimension fixed by its spectral exponent (Osborne & Provenzale, 1989; Theiler, 1986). The second regime is equally consequential and less often noticed: a decaying transient traces a curve, so its active dimension is 1 whatever the number of available directions.

3.4 ADMISSIBILITY DIAGNOSTICS

Both failure regimes can be detected without a ground truth, by two statistics and one flag. The **identifiability ratio** $\rho_{\text{ident}} = \hat{d}_{\text{MG}}(2E_{\text{max}})/\hat{d}_{\text{MG}}(E_{\text{max}})$ is near unity when a dimension is resolvable and grows when the estimate is a property of the embedding space rather than of the data. The **recurrence count**, the number of times the series crosses its own local trend inside a window, identifies the transient regime, in which ρ_{ident} is near unity for the wrong reason. The **degeneracy flag** returns no value when the neighbour distances or the log-ratio sum fall below their numerical floors. Section 6.4 gives their reference values.

4 VALIDATION PROTOCOL

Delay-embedding statistics return a finite number on data satisfying none of their hypotheses. Six requirements are fixed in advance, and every result here is held to them.

1. *The ground truth is measured, not assumed*, from equation (2) on the recorded trajectory and increments.
2. *The estimator configuration is selected once and frozen*, on a grid of estimator parameters only, since a grid over system parameters changes the data when a configuration is chosen.
3. *The split is disjoint in seed and in rank*, since withholding seeds alone leaves the excitation geometry shared whenever it is a function of the rank.
4. *A zero learning rate must silence the observer*, which otherwise reads the external drive through the residual rather than through the optimiser.
5. *The three statistics of section 3.2 are reported on every call*, and a result in which one of them performs comparably is reported as such.
6. *Excitation bandwidth is matched across ranks*, or a construction in which each added direction also adds a higher frequency lets the roughness ratio alone order the rank and the experiment distinguishes nothing.

5 EVALUATION ON SYSTEMS WITH A KNOWN ACTIVE DIMENSION

The estimator is evaluated on five systems of increasing realism, from a prescribed oscillation with no optimiser to a network trained on image data. In each, r is fixed by construction and then verified by equation (2), and the estimator sees only a scalar series (table 2).

5.1 AN OSCILLATING DIAGONAL MATRIX

The first system has no optimiser and no feedback, which makes a negative result decisive and a positive one weak. Let $\mathbf{W}(t) = \text{diag}(b_i + \delta_i(t))$ with $b_i \neq 0$ and exactly r coordinates oscillating at rationally independent frequencies inside one octave. The orbit closure is an r -torus, so the active dimension is exactly r ; the offsets must be non-zero or the squared Frobenius norm is not injective on it. One configuration was frozen on a single seed, and two further seeds and a non-monotone schedule of ranks were withheld. Recovery on the withheld data is accurate to within half a component up to ten directions and saturates above about nine when the range is extended to twenty.

This system also supplies the control that decides whether the estimate reflects geometry at all. Driving r coordinates from one phase with distinct fixed offsets gives r visibly moving coordinates and a single degree of freedom; the estimate separates that case from four independent oscillators by a factor of 3.4, where the roughness ratio separates them by 5%. That is the clearest evidence available that the geometry does work smoothness cannot. On the same data LB carries a bias of +1.02 against -0.08 for MG, which is the $(m-1)/(m-2)$ correction of section 3.2 and nothing else.

5.2 ONLINE LINEAR AND LOGISTIC REGRESSION

The second and third systems introduce an optimiser. A $D = 64$ linear model, and then a logistic model trained with binary cross-entropy, are fitted by full-batch gradient descent on one-hot inputs, with exactly r target coordinates moving quasiperiodically and the rest constant and already fitted, so the active dynamics have dimension r . The best observers, the parameter Frobenius norm in the linear model and the output probability norm in the logistic one, both order the rank almost perfectly to twenty and both saturate above about twelve directions, which places the saturation point of section 5.1 in a system with feedback and a nonlinear link. Their configuration was selected in breach of requirement 2, so the errors in table 2 locate that point but do not measure recovery.

5.3 A FROZEN NONLINEAR DECODER AND A CONSTRAINED PERCEPTRON

The fourth and fifth systems add nonlinearity and backpropagation: a trainable $\mathbf{z} \in \mathbb{R}^r$ optimised through a frozen nonlinear decoder whose linear skip connection keeps the map full rank, and a two-layer perceptron of $P = 243$ parameters restricted to $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{U}\mathbf{z}$ with $\mathbf{U}$ orthonormal and only $\mathbf{z}$ trained, so that it has hundreds of parameters and exactly r available directions.

Read naively both succeed, and the numerical rank of the map from $\mathbf{z}$ to the predictions equals r in every withheld run. Both nevertheless fail requirement 4: with the learning rate set to zero, so that no training occurs, the perceptron experiment reproduces its reported result almost exactly, the two series correlating above 0.98 at every rank. The drive acts on the targets and therefore reaches the observers through the residual, without passing through the optimiser. Section 5.4 rebuilds the design with the drive acting on the loss weights instead, keeping the nonlinearity and the backpropagation that motivate these two systems and passing the control they fail.

5.4 A NETWORK TRAINED ON IMAGE DATA

The fifth system is the parameterisation of section 5.3 on real inputs. A tanh perceptron $64 \rightarrow 96 \rightarrow 96$ is trained on the sklearn digits dataset and frozen, and a linear head on top of it is confined to a k -dimensional affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{V}^\top \mathbf{c}$. The loss is softmax cross-entropy, so the curvature is data-dependent and anisotropic; all dynamics occur in $\mathbf{c}$ and the logs are scalar functions of it. The excitation is made r -dimensional rather than nominally so by modulating the loss weights of r of twelve fixed data groups with rationally independent sinusoids and orthogonalising the resulting gradient directions (Appendix E), after which the measured active dimension equals r to within 0.008

Table 2: The five systems, ordered by increasing realism. “Truth” states whether the active dimension is fixed by construction alone or additionally verified by equation (2); only rows marked complete satisfy every requirement of section 4.

system	truth	range	MAE	ρ	protocol
oscillating matrix	constructed	1–10	0.30 / 0.69	0.99–1.00	complete
oscillating matrix	constructed	1–20	1.37	0.95–0.99	ordering only
online linear regression	constructed	1–20	1.34	0.980	ordering only
logistic regression	constructed	1–20	1.33	0.986	ordering only
frozen nonlinear decoder	constructed	1–20	2.13	0.924	fails req. 4
perceptron in a k -subspace	constructed	1–20	2.06	0.939	fails req. 4
image data, parameter subspace	measured	1–8	0.87	+1.000	complete
image data, parameter subspace	measured	1–20	1.62	+0.973	rank split only
image data, function subspace	measured	1–8	0.50	+0.988	complete

components over the whole range. Six excitation regimes are run: two deterministic and recurrent, at a fast and a slow drive period; injected rank- r gradient noise; real mini-batch noise projected to rank r ; plain mini-batch descent; and a decaying transient. The last two are results rather than settings, since the active dimension of a transient is one whatever r is, and the noise rank of ordinary mini-batch descent is fixed by the gradient covariance of the data.

Recovery. Twelve observers were used, in five families: loss, norms, gradient, fixed random projections of the parameters, and function-space quantities. The estimator’s five parameters were selected once, on seeds and ranks reserved for that purpose, and frozen. On the withheld ranks and seeds, over the ten usable state-only observers, the mean absolute error in the deterministic recurrent regime is 0.87 components taken run by run, and a median of 0.47 when the four seeds at each rank are averaged first, at a median slope of 0.92 and a perfect rank correlation for every one of the ten. Errors by observer are in Appendix C; the parameter norms, which are what a run ordinarily logs, sit in the middle of the range, and are also the two observers whose roughness is not flat against the rank, so part of their ordering may be spectral rather than geometric. The instantaneous mini-batch loss is excluded from every aggregate despite scoring among the best three, because it alone fails requirement 4: frozen at zero learning rate, with the measured active dimension identically one at every rank, its estimate still rises with the nominal rank, since it contains the loss weights and therefore the drive.

Twenty directions. Repeating the design with twenty available directions gives the same picture with a lower ceiling: recovery is close to exact to about six directions and then saturates, reaching 15.1 at a true 20.0, while the ordering survives throughout at Spearman +0.97. Above about eight directions the spread across seeds is comparable to the error, so the estimate is an ordinal score rather than a count, and that configuration was scored on the seeds it was selected on, so its split satisfies requirement 3 in rank only. Appendix C gives the per-rank values and the two comparisons that matter: MG is behind the uncorrected average LB at both ranges, and its advantage over PR_{delay} is a factor of about two at eight directions and about 1.4 at twenty. A second construction, defining the subspace in function space instead (Appendix E), is the most accurate row of table 2, but its rank is imposed by an external teacher.

5.5 SUMMARY

Only the image-data rows of table 2 carry a recovery error that may be read as one, the rows failing requirement 4 being listed so that the control rejecting them is documented. The picture is consistent across five systems and two ranges: recovery of the absolute value holds to about six components and degrades gradually above it, the ordering holds throughout, and the error on a real network with a real optimiser is of the same order as on a prescribed oscillation with no optimiser.

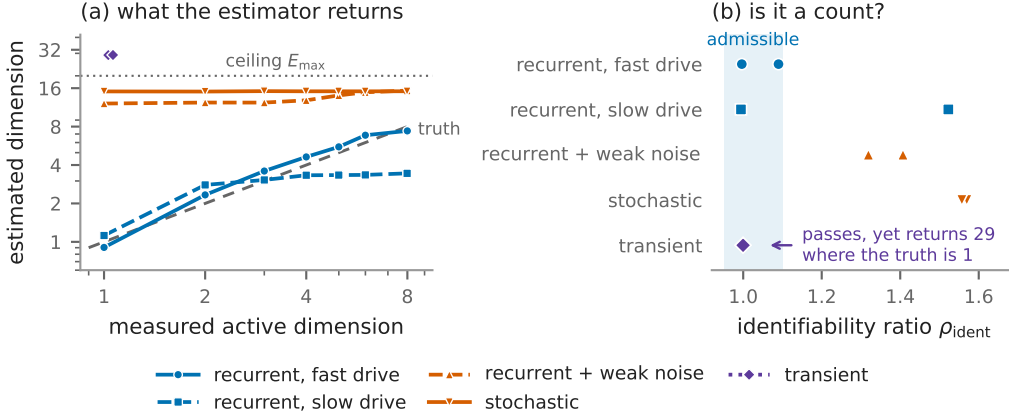


Figure 1: The evaluation of section 5.4: a tanh network trained on sklearn digits and frozen, a linear head confined to ten directions of which r are excited, five excitation regimes. (a) The estimate against the measured active dimension, median over four seeds and ten observers; it tracks the truth only under a fast deterministic recurrent drive. (b) The identifiability ratio on the same runs, which separates those regimes without a ground truth. The transient is the one regime it passes, hence the recurrence count of section 3.4.

6 CONDITIONS OF VALIDITY

6.1 DYNAMICAL REGIME

Figure 1 evaluates the estimator in each regime of table 1 at the same nominal rank, observer and matched octave. Under stochastic excitation it is flat in the rank, responding to $E_{\max}$, the correlation time and the record length rather than to the truth, and the error exceeds ten components. Weak noise added to a recurrent trajectory destroys the absolute value while leaving the ordering: at a quarter of the pure noise amplitude, injected in the same r directions and leaving the measured active dimension exactly r , the error rises tenfold while the Spearman correlation stays above $+0.96$. A training log is a weak deterministic signal inside stochastic gradient fluctuation, so that is the operative regime in section 7. A decaying transient returns a large reproducible estimate above 27 components where the truth is one, and is the one regime in which ρ_{ident} does not fire.

6.2 THE DELAY LAG

The delay window spans $(E_{\max} - 1)\tau$ samples and must cover a real fraction of the oscillation period, or the delay coordinates are collinear and the torus does not unfold. On a fixed torus of period 400 at eight active directions the estimate reads 2.60 when the window spans a twentieth of a period, 8.53 at four tenths and 63.35 at one and a half: with the system unchanged, τ alone moves it by more than sixty components. The correct lag is a function of the oscillation period, which a controlled experiment knows and a training log does not; section 7 estimates it from the autocorrelation time.

6.3 NUISANCE FACTORS

At a fixed active dimension of four we vary seven factors that leave it unchanged, over five seeds and twelve observers, and compare the shift against a real four-component change; the seed-to-seed variability of an unchanged configuration is 0.21 components, a twentieth of that reference (Appendix D). The smoothness control is the decisive one and it passes: moving the drive band by an octave changes the autocorrelation time and the roughness fourfold, yet moves the estimate only to the level of the baseline. A constant rescaling of the observer leaves it exactly invariant, verified to machine precision at twenty directions; a *ramped* gain does not, and moves it by about a third of a real four-component change. A run whose parameter norm grows can therefore drift by order one component with nothing dimensional occurring, which is the nuisance factor that matters most in section 7.

Table 3: The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Runs that have not generalised within the budget are censored, not negative.

run	generalises	MG	ρ_{ident}	roughness	PR_{delay}	crossings
transformer, positive control	yes	15.31	1.289	0.030	n/a	56
transformer, reduced data, s0	yes	4.93	1.599	0.068	n/a	146
transformer, reduced data, s2	no	4.47	1.514	0.149	n/a	151
transformer, no weight decay	no	28.08	0.999	0.000	n/a	2
perceptron, $n + m$	yes	19.54	0.999	0.001	1.0	n/a
perceptron, $n \cdot m$	yes	19.63	0.999	0.001	1.0	n/a
perceptron, $n^3 + nm^2 + m$	no	22.43	0.998	0.000	1.0	n/a

6.4 VALIDATION OF THE DIAGNOSTICS

Using no ground truth, the identifiability ratio separates the regimes in which the estimate is accurate, where it lies between 1.00 and 1.09, from those in which it is not, where it lies between 1.32 and 1.57; under the slow drive it separates within a regime too, reading 0.99 where the estimate is correct and 1.52 where it has saturated. It is not sufficient alone, since it reads 1.00 on the transient regime, and must be paired with the recurrence count, which is near zero there. The degeneracy flag fires on eleven of the thirty-five run-by-column cells of the logs of section 7.

7 APPLICATION TO GROKING

Grokking (Power et al., 2022) is studied in two experimental settings that differ in every relevant respect, and we reproduce both (every run is listed in Appendix J): the regularised stochastic one of Liu et al. (2023), a one-layer transformer trained on modular addition and on composition in S_5 with AdamW, weight decay 1.0, cross-entropy and mini-batches; and the unregularised full-batch one of Gromov (2023) and Doshi et al. (2024), a two-layer perceptron with quadratic activation trained on mean squared error. The second supplies label-matched pairs of targets differing by one monomial, of which one generalises and the other does not while both fit the training set completely (Appendix I).

7.1 REGIME CLASSIFICATION OF THE TRAINING LOGS

Applying the frozen configuration of section 5.4 gives table 3, which figure 3 plots: the two settings occupy different regimes of table 1, and the diagnostics report which without reference to any label. The regularised stochastic runs sit in a band of the identifiability ratio the deterministic regimes never reach, with the roughness and recurrence count of a noisy series, and their estimates span a factor of four across seven runs of one configuration family: what figure 1 predicts for a stochastically excited series, whose value is set by $E_{\max}$, the correlation time and the record length rather than by a rank. The unregularised full-batch runs sit instead at an identifiability ratio of unity with a roughness of one part in a thousand, two trend crossings per window and a delay participation ratio of exactly one: the signature of the transient regime, where figure 1 records an estimate above 27 against a true active dimension of one, and consistently with that the values here sit at or above the ceiling $E_{\max} = 20$. The identifiability ratio alone would pass them; the recurrence count is what rejects them. In both settings the absolute value is therefore not a count of components, and the diagnostics identify the reason in each case: the method reports the regime before it reports a number.

7.2 SEPARATION OF LABEL-MATCHED PAIRS

The perceptron setting supplies controls the first cannot construct: both members of a pair reach full training accuracy under identical optimisation on identical data budgets and differ by one monomial in the target, so anything separating their logs is a property of the solution found. On the training loss the estimate is about three units higher for the non-generalising member in all four pairs, with no overlap, from a single scalar log (Appendix F). It is not, however, a dimension: table 3 places all six runs in the transient regime, where the estimate follows the shape of the curve rather than any attractor, and the mechanism is in the standard log, since the representable targets drive the training

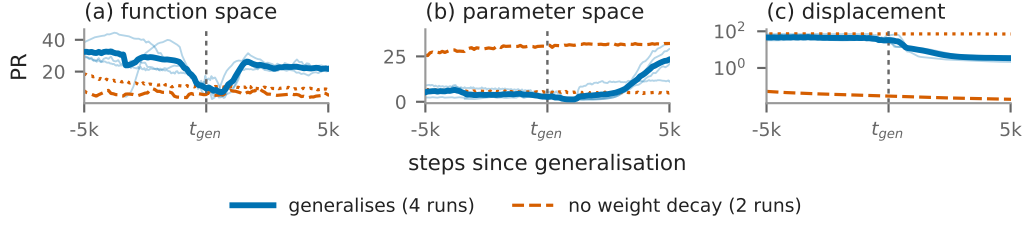


Figure 2: The active dimension measured directly, on six transformer runs: three seeds of modular addition and one S_5 composition at weight decay 1.0, against the same configurations at weight decay zero, which never generalise. Sliding windows of 600 steps, aligned on each run’s own generalisation step; thick blue is the mean of the four generalising runs.

error to zero while the perturbed ones stop three orders of magnitude short despite full training accuracy, fitting the argmax but not the regression.

7.3 DIRECT MEASUREMENT OF THE ACTIVE DIMENSION

The active dimension of equation (2) needs no delay reconstruction when the trajectory itself can be stored. We store a sketch of it, in parameter space and in function space, at every logged step of a transformer trained to grok (Appendix G). At generalisation the trajectory passes through a transient one-dimensional bottleneck (figure 2): in function space the participation ratio falls by a factor of five to twelve, in parameter space to between 1.09 and 1.60 some six hundred to a thousand steps later, and then re-expands well above its plateau. Three observations support the reading: it sits at generalisation in seeds whose generalisation step differs by a factor of 3.7 and in a second task, so it is not a function of training progress; the displacement falls as well but reaches its floor three thousand steps later and never recovers while the rank does, so it is not the trajectory coming to rest; and the runs without weight decay do not show it.

The same measurement on the perceptron setting returns nothing, for a reason that is a resolution limit rather than a result: over a 600-step window a deterministic trajectory is straight to within the measurement precision, so the statistic is pinned near unity. Lengthening the window to 12 000 steps gives it range, and what appears is a maximum at memorisation that the non-generalising control shares (Appendix H).

7.4 WHAT FALLS, AND WHEN

A fall of the estimate is not specific to generalisation. In a 200 000-step run at $p = 211$ with no weight decay, which memorises at step 2 250, ends at 3.6 % validation accuracy and grows its parameter norm from 43 to 126, a fall is detected at step 49 250, for which the amplitude control of section 6.3 gives the mechanism. Nor is the absence of generalisation within a budget evidence that a fall was a false alarm: two reduced-data configurations against which earlier specificity figures were computed generalise once the budget is extended, at steps 39 600 and 110 940. The picture is consistent with section 7.3: the trajectory simplifies sharply and transiently at generalisation and also simplifies elsewhere, so a fall is evidence of a change of regime rather than of generalisation.

8 LIMITATIONS AND CONCLUSION

One scalar log of a suitable observer recovers the number of directions an optimiser excites within a window to under one component up to eight directions, and orders it to twenty. What makes that number usable is not the estimator but the diagnostics beside it, which decide without a ground truth whether it counts components; applied to grokking they place both standard settings outside the admissible regime. Measured directly, the trajectory simplifies sharply at generalisation and also in runs that never generalise. Three limits bound this: the absolute value requires the deterministic recurrent regime and a lag matched to the oscillation period, it is ordinal rather than a count above eight directions, and the measurement of section 7.3 is a signature at generalisation rather than an early warning, and is not reproduced in the second setting.

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A THE ESTIMATOR

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ ; embedding dimension E ; neighbours m ; Theiler window W

```

1:  $z \leftarrow (x - \text{mean } x) / \text{std } x$ 
2:  $\mathbf{y}_t \leftarrow (z_t, z_{t-\tau}, \dots, z_{t-(E-1)\tau})$  for  $t = (E-1)\tau + 1, \dots, n$ 
3:  $S \leftarrow 0, N \leftarrow 0$ 
4: for each  $\mathbf{y}_i$  do
5:    $r_1 \leq \dots \leq r_m \leftarrow$  distances to the  $m$  nearest  $\mathbf{y}_j$  with  $|i - j| > W$ 
6:   if  $r_1 < \varepsilon_{\text{dist}}$  or  $\sum_j \log(r_m/r_j) < \varepsilon_{\text{ratio}}$  then
7:     mark window degenerate; return undefined
8:   end if
9:    $S \leftarrow S + \sum_{j=1}^{m-1} \log(r_m/r_j); \quad N \leftarrow N + 1$ 
10: end for
11: return  $N(m-1)/S$ 

```

The Theiler window W is set from the autocorrelation time of the series and capped. No clamping of the returned value to E is applied: a divergent estimate is reported as divergent, since clamping converts the transient regime of table 1 into a plausible number.

B FROZEN CONFIGURATIONS

Two estimator configurations are used in this paper, one for each range of the active dimension. Both were selected once, on data withheld from every later experiment, and then frozen. Table 4 states them and the grids they came from.

Table 4: The two frozen configurations and their selection. Only the swept parameters were free; the remainder were fixed before any estimate was computed.

	eight-direction	twenty-direction
E_{max}	20	40
τ	4	16
m (neighbours)	20	20
Theiler window	autocorrelation	autocorrelation
window / stride	8 000 / 2 000	8 000 / 4 000
dither	10^{-9}	10^{-9}
swept	$E_{\text{max}}, \tau, m, \text{Theiler, window}$	E_{max}, τ
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–4	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.324	2.47
selection error, worst	1.499	3.35

The selection objective in both cases is the error of the median estimate against the median measured active dimension, over the four observers reserved for selection, one from each observer family. The eight-direction objective adds a penalty on the rank correlation and on the spread across seeds; at the twenty-direction configuration both penalties are identically zero, so the objective there reduces to the error alone.

Four properties of these choices should be stated with them.

Both sit at a grid boundary. The eight-direction configuration is at the top of all five swept axes and the twenty-direction one at the top of both. That is less alarming than it sounds for the first, since four of its five axes have only two levels, so every configuration in that grid is at a boundary in four of

them; τ is the only axis with an interior value, and the winner is at its top end. It does mean neither configuration is interior, and that a wider grid could move the level.

The selection error spans a factor of five on identical data. Across the eight-direction grid the error runs from 0.324 to 1.499 on the same logs, so the level is a choice as much as a measurement, and only the withheld numbers of section 5.4 may be read as recovery.

The Theiler axis is inert on this data. The autocorrelation rule returns the larger of the embedding span $(E_{\max} - 1)\tau$ and the measured autocorrelation time. On these logs the autocorrelation time is one to three samples while the embedding span is nine or more, so the two settings return bit-identical values at all 24 points of the grid where they are compared. The 48-point grid therefore contains 24 distinct measurements, and the choice of Theiler rule is not a choice at all here.

The twenty-direction configuration does not use the window its parameters name. At $E_{\max} = 40$ and $\tau = 16$ the requested exclusion is 624 samples, which the implementation truncates to its cap of 150 for the cost reason given in Appendix A. The eight-direction configuration is uncapped.

C RESULTS BY OBSERVER, AND THE ALTERNATIVES

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.4, in the deterministic recurrent regime, and are scored against the measured active dimension rather than against r . Where an aggregation over seeds is needed, the median over the four seeds at each rank is taken before scoring; that is the convention the analysis uses throughout, and it is stated because the choice moves the numbers. Table 5 gives the result for every observer.

Table 5: Per-observer results on withheld ranks and seeds. The last column is the rank correlation between the roughness statistic and the measured active dimension: a value near zero means the ordering that observer produces is not available from smoothness alone.

observer	family	MAE	slope	ρ	$\rho(\text{roughness})$
probe loss	loss	0.303	0.904	+1.000	+0.200
function-space norm	norm	0.361	0.931	+1.000	+0.200
margin	function	0.409	0.931	+1.000	+0.200
parameter norms (two)	norm	0.448	0.849	+1.000	+1.000
fixed parameter projection	projection	0.489	0.966	+1.000	+0.200
gradient projection	gradient	0.559	0.891	+1.000	+0.800
function-space projection	function	0.699	0.705	+1.000	-0.200
gradient norm	gradient	1.972	1.654	+1.000	-0.800
full-batch loss	loss	2.008	1.621	+1.000	-0.800
instantaneous loss	loss	0.418	0.863	+1.000	0.000
probe accuracy	function	degenerate in every window			

Two observers are excluded from every aggregate, for different reasons. Probe accuracy is quantised, which drives the delay vectors together and trips the degeneracy flag in 100 % of its windows; it is the sole source of the degenerate fraction reported anywhere in this study, every other observer being at zero. The instantaneous mini-batch loss is excluded because it fails requirement 4: at zero learning rate, where the measured active dimension is exactly one at every rank, its estimate still reads 0.91, 2.22, 3.71, 4.62, 5.45, 5.31 and 6.93 across $r = 1, \dots, 8$, a range of six components produced entirely by the drive. The other eleven observers are exactly constant under that control, so no estimate is defined for them.

The roughness column carries a caveat that the main text states and that this table makes precise. For seven of the nine usable observers the roughness statistic carries little or no ordering, so their agreement with the truth is not available from smoothness. For the two parameter norms it orders the withheld ranks perfectly, and those are exactly the observers a training run ordinarily logs. Their result should therefore be read as consistent with, rather than as evidence against, a smoothness explanation.

Table 6: Twenty available directions, image data, median over three seeds and eight observers. The truth is the measured active dimension; the last row is the spread of MG across seeds at fixed rank.

true r	1	2	4	6	8	10	12	14	17	20
MG	0.89	2.35	4.41	6.66	7.38	8.99	10.61	11.40	13.50	15.09
PR _{delay}	2.00	2.29	4.54	6.81	7.95	9.20	10.79	12.15	11.71	13.51
spectral PR	1.01	1.25	2.58	3.68	4.18	5.48	5.98	6.53	8.11	8.38
spread across seeds	0.00	0.20	1.49	2.15	2.11	2.95	1.50	2.84	2.76	3.33

Table 7: The estimator against the alternatives, on the same data and under the same convention, at both ranges. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum.

statistic	eight directions		twenty directions	
	MAE	ρ	MAE	ρ
MG †	0.457	+1.000	1.632	+0.961
LB †	0.435	+1.000	1.358	+0.979
TwoNN †	1.814	+1.000	3.296	+0.864
delay participation ratio	0.894	+1.000	2.320	+0.982
spectral PR, 256 bands	1.604	+1.000	5.401	+0.996
spectral PR, 1024 bands	1.353	+1.000	4.547	+0.996
spectral PR, native resolution	1.026	+0.800	2.760	+0.968
roughness	3.652	+0.200	10.069	-0.482

Table 7 should be read with three things in mind, and none of them favours MG.

First, MG is not the most accurate statistic at either range. The uncorrected per-point average is ahead of it at both, by 0.02 components at eight directions and by 0.27 at twenty. The correction of section 3.2 removes a positive bias, and on a saturating estimate that bias was partly cancelling the saturation, so removing it costs accuracy exactly where the estimate is already saturating.

Second, MG is not the best-ordering statistic at twenty directions. The spectral participation ratio at 256 and 1024 bands orders the ranks more faithfully than MG does, at four times the error. If only the ordering is wanted, a statistic requiring no embedding and no neighbour search will supply it.

Third, MG’s advantage over the linear delay participation ratio is a factor of about two at eight directions, not the factor of six that the selection split of Appendix B reports. That larger figure is computed on the ranks and seeds the configuration was chosen on and is a property of the selection, not a recovery result.

The advantage depends on the observer. Running the same torus through two scalar observers that differ only in how evenly they weight the modes settles which statistic is preferable, and the answer is that it depends. Under an observer that weights all modes equally, the spectral participation ratio at 256 bands attains an error of 0.35 against MG’s 1.19, ahead by 3.4 times. Under a generic observer with random weights, which is what a random projection of a network’s parameters actually is, MG attains 0.34 against 1.61, ahead by 4.7 times. Both orderings are perfect in three of those four cells, so what separates the statistics is absolute recovery and not rank agreement. Both figures also use the delay lag that is best for that observer, which a controlled experiment can choose and a training log cannot: at a lag common to both observers the two factors become 6.6 and 1.9 respectively.

D CONDITIONS OF VALIDITY IN DETAIL

The delay lag. The delay window spans $(E_{\max} - 1)\tau$ samples, and table 8 sweeps τ on a fixed torus of period 400 at $E_{\max} = 20$, leaving the system unchanged. At a span of one twentieth of a period every rank returns the same tangent constant near 2.5; at four tenths the estimate follows the rank to within one component; at one and a half periods it exceeds the rank eightfold. The estimate at

$r = 8$ moves by 61.5 components across the sweep and the estimate at $r = 1$ by 0.03, so the failure is not a uniform bias but a collapse of the ordering at one end and a divergence at the other.

Table 8: Estimate against the delay lag on a fixed torus of period 400, $E_{\max} = 20$. The span is the delay window in units of the oscillation period. The correct rank is the column heading.

τ	span	$r = 1$	2	3	4	6	8
1	0.05	1.42	2.44	2.49	2.63	2.44	2.60
4	0.19	1.43	2.72	3.31	3.83	3.94	4.29
8	0.38	1.43	2.77	4.18	6.16	6.57	8.53
16	0.76	1.44	3.00	5.92	13.66	12.83	20.76
32	1.52	1.46	3.06	9.90	33.28	31.02	63.35

Nuisance factors. Table 9 varies seven factors at a fixed active dimension of four, over five seeds and twelve observers, against the real four-component change measured below. Rotating the coordinates is scored on the three projection observers alone, since the remaining nine are invariant to it by construction, so it measures a property of the observer rather than of the estimator.

Table 9: Shift in the estimate produced by factors that leave the active dimension unchanged. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale against which the others should be read.

factor	$ \Delta \text{MG} $	as % of a real four-component change
none (baseline)	0.21	5
learning rate halved	0.16	4
drive band up one octave	0.49	12
drive band down one octave	0.50	13
coordinates rotated	0.86	22
observer gain ramped $10\times$	1.16	29
state amplitude ramped $4\times$	1.32	33

Detecting a change. Recovery of a static level does not establish that a change is detected. Running three consecutive segments $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$ with nothing else altered, and re-measuring the ground truth on each segment, a true fall of 3.99 components is observed as 3.95 in the deterministic recurrent regime, is detected by every usable observer in both directions, and recovers to the level before it to within 0.01 components. Under stochastic excitation the same true change is observed as 0.15.

The rule fires when the estimate falls below the median of the pre-switch segment by three times that segment’s window-to-window scatter, for two consecutive windows. The threshold uses the pre-switch segment alone, never the midpoint of the two observed levels, which would hand the detector the answer and would decide the arm where the two levels coincide by a coin flip. On the parameter norm the level pairs read $4.77 \rightarrow 1.22 \rightarrow 4.78$ for a true $4 \rightarrow 1 \rightarrow 4$ and $6.90 \rightarrow 2.12 \rightarrow 6.91$ for a true $6 \rightarrow 2 \rightarrow 6$; at $8 \rightarrow 3 \rightarrow 8$ they read $6.97 \rightarrow 3.64 \rightarrow 6.97$, under-reporting the high level by a component, which is the saturation of table 6 seen from another direction. The level error is 0.89 at the high level and 0.22 at the low one: the estimate is more accurate the fewer directions are active.

The response time cannot be quoted from this experiment. With a window of 8 000 samples and a stride of 2 000 the two observed lags, 999 and 1 999 steps, are the two floors of the measurement grid rather than measured values. What the experiment does establish is that the estimate crosses the threshold on a window still containing 87.5 % pre-switch data.

E SYSTEM CONSTRUCTIONS

The image-data system of section 5.4. The data is partitioned into twelve fixed groups, the same partition at every rank so that the partition is not itself a function of r , and the loss weights of r of them are modulated by r rationally independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j . These directions are neither orthogonal nor of equal effect, since

random data groups have correlated gradients, so a mixing matrix $\text{pinv}(\Phi)Q$ with Q an orthonormal basis of $\text{range}(\Phi)$ orthogonalises them, and the per-mode scalar response gain of the linearised update is divided out. Without this correction the trajectory participation ratio falls well below r .

Table 10 verifies that the construction achieves what it intends, and two of its rows are results rather than settings. The active dimension of a decaying transient is one whatever r is, since a converging trajectory is a curve; and plain mini-batch descent sits at 6.2 regardless of r , because its noise rank is fixed by the per-example gradient covariance of the data and is not a free parameter of the experiment. Projecting real mini-batch noise onto r directions recovers only part of the intended rank, reaching 5.6 at $r = 8$, which is why that regime is scored against its measured value and not against r .

Table 10: Measured trajectory participation ratio by excitation regime and nominal rank, median over seeds, for the ten available directions of section 5.4. Every estimate in this paper is scored against these values, never against r .

regime	$r = 1$	2	4	6	8	what is excited
qp	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast drive
qp_slow	1.00	2.00	4.00	6.00	7.99	the same at a training-log timescale
qp_nopre	1.00	1.99	3.99	5.89	7.55	qp without the preconditioner
mixed	1.00	2.00	4.00	6.00	7.99	qp plus noise in the same directions
noise	1.00	2.00	3.99	5.99	7.98	injected rank- r gradient noise
noise_nopre	1.10	2.08	3.74	5.47	6.43	the same without the preconditioner
batch_proj	1.00	1.79	3.25	4.53	5.61	real mini-batch noise projected to rank r
batch	6.20	6.25	6.21	6.26	6.20	plain mini-batch descent
gd	1.05	1.05	1.05	1.03	1.06	a decaying transient
qp, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	requirement 4 control

The last row is the control of requirement 4. With the learning rate multiplied by zero the active dimension is identically one at every nominal rank, so an observer whose estimate still tracks r there is reading the drive rather than the optimiser.

The functional rank is 10 at every rank, seed and regime, with a participation ratio between 8.40 and 9.03 across the four seeds; it is a property of the frozen backbone and does not depend on the excitation. The condition number of the drive-direction matrix is 1.00 at $r = 1$ by construction and has a median of 13.3 at $r = 8$, which is the anisotropy the mixing matrix exists to remove.

The function-space construction. A local adapter is built around a trained $64 \rightarrow 16 \rightarrow 10$ network and its function-space Jacobian is QR-whitened, so that its first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it by exact backpropagation through both layers. The functional, trajectory-covariance and update-covariance ranks were all verified to equal r .

The twenty-direction system. The same construction at twenty available directions reproduces the nominal rank to within 0.008 components over $r = 1, \dots, 20$ in the recurrent regime, and to three decimals up to $r = 10$. The other regimes behave as at ten directions: injected noise tracks r closely, projected mini-batch noise reaches only 9.4 at $r = 20$, and the decaying transient stays between 1.02 and 1.09 throughout.

F FIGURES FOR THE GROKING APPLICATION

Figure 3 places every training log of section 7 in the plane of the two diagnostics. An estimate is a count of components only inside the shaded rectangle, and no run of either setting is there: the regularised mini-batch runs are unstable in the embedding dimension, and the full-batch and zero-weight-decay runs are stable only because their series is monotone.

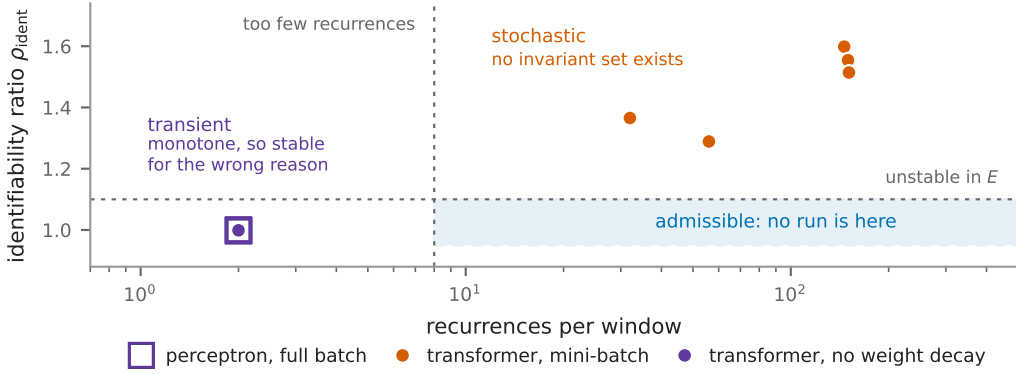


Figure 3: Every training log of this section in the plane of the two diagnostics of section 3.4, one point per run: seven transformer runs of 120 000 steps observed through the parameter norm, and ten perceptron runs of 100 000 steps observed through the training loss. The vertical axis is the identifiability ratio, near unity when the estimate does not depend on the embedding dimension; the horizontal axis counts how often the series crosses its own local trend inside a window. An estimate is a count of components only inside the shaded rectangle, and no run of either setting is there. The two settings fail in different corners, and both failures are visible without any label.

Figure 4 shows the separation of section 7.2 and the mechanism behind it. The estimate is about three units higher for the member of each pair that never generalises, in all four pairs and with no overlap, but the training loss shows that the two members differ in the shape of their curves long before any question of dimension arises.

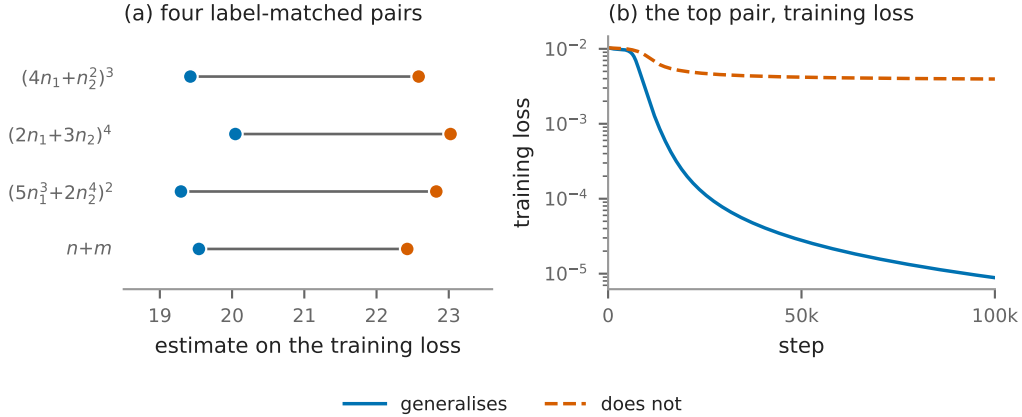


Figure 4: (a) The estimate on the training loss for the four label-matched pairs of section 7.2: three modular polynomials at $p = 97$, each against itself plus one monomial, and modular addition against $n^3 + nm^2 + m$. Both members of a pair reach full training accuracy under identical optimisation on identical data budgets. (b) The top pair of (a), and what the estimate is reading in it. The representable target drives the training loss to zero, while the perturbed one stops three orders of magnitude short despite fitting the training set completely, so the two curves differ in shape rather than in any attractor dimension.

G THE TRAJECTORY SKETCH

The direct measurement of section 7.3 needs the trajectory itself, which for the transformer is 226 816 doubles at each of two to three thousand logged steps. Every logged step is therefore compressed by a CountSketch (Charikar et al., 2002) to $\mathbb{R}^{1024}$, with two independent hash families, which preserves

the Gram matrix and so the covariance spectrum to $O(\sqrt{\log T/D})$, about nine per cent here. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length. The normalisation is not cosmetic. The raw logit scale is mechanically coupled to weight decay, so a signal read off it could not be distinguished from a detector of weight decay.

Three checks stand behind the numbers. Pushing synthetic trajectories of known rank through the same sketch and the same participation ratio returns 1.00, 1.98, 4.46 and 8.63 for true ranks 1, 2, 5 and 10, so the compression does not manufacture or destroy rank in the range of interest. On a real run of the perceptron architecture, where the parameter count is small enough to allow it, the participation ratio computed exactly on all 145 500 raw parameters agrees with the sketched value to three decimals. And the observer is verified not to perturb training by requiring the log to be bit-identical with and without it, which matters because the task module seeds one global random stream that the train-validation split, the initialisation and the mini-batch order all continue: a single stray draw changes the initial weights and can destroy grokking.

Windows are sixty logged rows, which is 600 optimiser steps for the modular runs and 300 for S_5 , and are labelled by their centre so that the position of a feature is not an artefact of the labelling. Four statistics are computed per window in each space: the participation ratio of the positions after removing a linear trend, since a steady drift is rank one and would otherwise mask everything else; the participation ratio of the increments block-averaged over five logged steps, to suppress the mini-batch noise floor; and the total displacement, which is the null for a rank that falls simply because the trajectory stopped moving.

H THE FULL-BATCH SETTING

Repeating the direct measurement on the perceptron setting gives a result that must be read as a limit of the measurement rather than as a negative. Under full-batch gradient descent every window of every run has a participation ratio between 1.000 and 1.362. This is not an artefact of the sketch: computed exactly on all 145 500 raw parameters the value is the same to three decimals, and the sketch recovers synthetic trajectories of known rank up to ten. A deterministic update makes the trajectory a smooth curve, and over a 600-step window a smooth curve is straight to within the measurement precision, so the statistic reports the window length rather than the dynamics. It has no range in which a collapse could appear, and the absence of a feature is therefore not evidence of its absence: the window must be set in units of the curvature scale of the trajectory. Repeating the measurement against window length confirms the diagnosis and withdraws the opening it appeared to leave. Two runs of 150 000 steps, a generalising one and its label-matched control, are measured over windows spanning 600 to 120 000 optimiser steps, with the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound on the statistic and its noise floor stay exactly where the published measurement put them and only the span changes. Table 11 gives the result. The statistic does leave the floor, at a scale of about 10^4 steps, two orders of magnitude above the window that produced the 1.000–1.362: the absence of a feature there was a resolution limit and not a fact about the run.

It is not, however, a hidden signal. What appears at 12 000 steps is one localised maximum, and both runs have one, of the same size and in the same place: 1.877 centred at step 12 800 for the run that memorises at 9 200 and generalises at 11 950, and 1.843 centred at 9 350 for the control, which memorises at 9 800 and never generalises. The feature tracks memorisation, which both runs do. Function space agrees with more range and no more separation, 2.618 against 2.145. Beyond 30 000 steps the ordering reverses and the control moves further than the generalising run, because the window is then a large fraction of a run whose weight norm is still climbing and the statistic is reading that drift. The full-batch trajectory therefore carries no rank signature of generalisation at any window length between 600 and 120 000 steps, which is the stronger form of the same conclusion: the collapse of section 7.3 belongs to the regularised stochastic setting, and a trajectory participation ratio must be controlled for window length as well as for batch size before anything is read from it.

Table 11: Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window. Sixty samples per window at every length. The 600-step row is the measurement reported above, on the same configuration and seed.

window (steps)	generalises		control, never generalises	
	median	max	median	max
600	1.002	1.035	1.001	1.011
3 000	1.070	1.253	1.003	1.092
6 000	1.028	1.537	1.005	1.360
12 000	1.011	1.877	1.020	1.843
30 000	1.007	2.111	1.113	1.725
60 000	1.021	1.660	1.335	1.975
120 000	1.256	1.659	1.929	2.337

Introducing mini-batches confirms the diagnosis. The same statistic then ranges over more than a decade while the generalisation step moves by one logged step, so the learning is unchanged and the statistic has moved by an order of magnitude. Any claim read from a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

I THE REPRESENTATION DIMENSION IN CLOSED FORM

The perceptron setting of section 7 gives the dimension of the learned representation a closed form, since the generalising solution uses $(p + 1)/2$ Fourier modes and nothing else. The order parameter is the mean inverse participation ratio of the first layer’s Fourier spectrum, which is 1.000 for that solution because each neuron carries a single frequency, and about 0.041 at random initialisation. Two things follow from measuring it, and both qualify readings of the representation dimension elsewhere.

The representation is not complete at the transition. For modular addition the order parameter reads 0.067 just after memorisation at step 9 170, 0.189 just before generalisation at step 11 950, 0.263 when validation accuracy first reaches 100 %, and then rises for a further 87 000 steps to 0.595 against an analytic 1.000. It does not jump at the transition; it begins rising at memorisation and passes smoothly through. Generalisation therefore arrives well before the representation is complete, and requires it only to be periodic enough. The effective rank of the same layer is far weaker as a signal: it spans 132.6 to 150.9 over the run against an analytic 148.8, is not monotone, and overshoots the reference before falling back, so landing near the reference is not evidence.

The order parameter is only interpretable against its own reference. Table 12 gives the measured value beside each task’s own closed form. Two of the tasks that generalise have a reference near the floor, because the representation is periodic in $g_1(n)$ while the spectrum is taken over the raw index n , and g_1 is not linear there. Read against modular addition’s 1.000, a measured 0.044 says nothing was learned; read against that task’s own 0.062, it says the run has essentially converged. The tasks that never generalise sit at the initialisation floor, and the analysis returns no reference for them rather than substituting another task’s.

Table 12: Fourier inverse participation ratio of the first layer at the end of training, against each task’s own closed form. All runs are full-batch gradient descent with no weight decay at $p = 97$. A dash marks a task for which the source papers give no closed form.

task	generalises	measured	own reference	grokking step
$n + m$	yes	0.595	1.000	11 950
$n - m$	yes	0.582	1.000	11 570
$n^2 + m^2$	yes	0.093	0.052	10 380
$n \cdot m$	yes	0.073	—	11 600
$(4n_1 + n_2^2)^3$	yes	0.243	1.000	12 560
$(2n_1 + 3n_2)^4$	yes	0.283	1.000	7 750
$(5n_1^3 + 2n_2^4)^2$	yes	0.044	0.062	6 430
$(4n_1 + n_2^2)^3 + n_1n_2$	no	0.040	—	none
$(2n_1 + 3n_2)^4 - n_1^2$	no	0.043	—	none
$(5n_1^3 + 2n_2^4)^2 - n_2$	no	0.039	—	none
$n^3 + nm^2 + m$	no	0.040	—	none

J RUN INVENTORY

Table 13 lists every training run the paper draws on. Two architectures appear: **T** is the one-layer transformer of Liu et al. (2023), 226 816 parameters at $p = 113$, trained with AdamW at learning rate 10^{-3} and batch size 256; **P** is the two-layer quadratic perceptron of Gromov (2023), 145 500 parameters at $p = 97$ and width 500, trained by full-batch gradient descent with no regularisation. Memorisation and generalisation steps are the first logged step at which training and validation accuracy respectively reach 95 %.

Table 13: Training runs used in this paper. α is the training fraction. A dash in the generalisation column marks a run that had not generalised when its budget ended; such runs are censored observations, not established negatives.

run	task	arch	wd	α	budget	mem.	gen.	used in
<i>Trajectory sketch attached</i>								
mod_wd1	mod. add. $p=113$	T	1.0	0.30	20k	1 470	13 700	section 7.3
mod_wd1_s43	mod. add. $p=113$	T	1.0	0.30	20k	1 580	6 660	section 7.3
mod_wd1_s44	mod. add. $p=113$	T	1.0	0.30	20k	1 410	3 680	section 7.3
s5_wd1	S_5 composition	T	0.2	0.50	15k	1 280	6 735	section 7.3
mod_wd0	mod. add. $p=113$	T	0.0	0.30	20k	630	—	section 7.3
s5_wd0	S_5 composition	T	0.0	0.50	15k	1 015	—	section 7.3
<i>Extended reruns, 120 000 steps</i>								
gropos_s0	mod. add. $p=113$	T	1.0	0.30	120k	1 450	5 110	section 7.1
lowdata15_s0	mod. add. $p=113$	T	1.0	0.15	120k	620	110 940	section 7.4
lowdata15_s1	mod. add. $p=113$	T	1.0	0.15	120k	630	—	section 7.4
lowdata15_s2	mod. add. $p=113$	T	1.0	0.15	120k	660	—	section 7.1
lowdata20_s0	mod. add. $p=113$	T	1.0	0.20	120k	800	39 600	section 7.4
wd0_s0	mod. add. $p=113$	T	0.0	0.30	120k	610	—	section 7.1
wd0_s1	mod. add. $p=113$	T	0.0	0.30	120k	630	—	section 7.1
p211_wd0	mod. add. $p=211$	T	0.0	0.16	200k	2 250	—	section 7.4
<i>Perceptron, $p = 97$, full batch, no regularisation</i>								
a_add	$n + m$	P	0	0.5	100k	9 170	11 950	section 7.2
a_mul	$n \cdot m$	P	0	0.5	100k	9 130	11 600	section 7.1
x_no_grok	$n^3 + nm^2 + m$	P	0	0.5	100k	9 760	—	section 7.2
g_p1	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7 410	12 560	section 7.2
g_p1x	$+ n_1 n_2$	P	0	0.5	100k	10 050	—	section 7.2
g_p2	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6 520	7 750	section 7.2
g_p2x	$- n_1^2$	P	0	0.5	100k	9 780	—	section 7.2
g_p3	$(5n_1^3 + 2n_2^4)^2$	P	0	0.5	100k	5 880	6 430	section 7.2
g_p3x	$- n_2$	P	0	0.5	100k	9 430	—	section 7.2

Four further runs repeat the top four perceptron rows with the trajectory sketch attached, at full batch and again with mini-batches of 512, for the measurement of Appendix H; a third pair repeats them at $p = 23$. Their memorisation and generalisation steps agree with the rows above to within one or two logged steps, which is what makes the comparison of Appendix H a comparison of noise rather than of learning.

Three notes on how the censored rows should be read. lowdata15_s1 ends its budget at 94% validation accuracy and still rising, so it is the clearest censored observation in the set rather than a negative. The generalisation step of lowdata15_s0 is 110 940 under the sustained-block criterion used for the extended reruns and 109 860 under the plain first-crossing rule used elsewhere; nothing in the paper depends on the difference. And g_p3x reaches 73% validation accuracy against a majority-class baseline of 1.3% while being provably outside the class the architecture can represent, so it is the one row in this table that neither of the source papers accounts for.